

RevenuePulse

Performance Case Study — SaaS Revenue Analytics Dashboard

OVERVIEW

RevenuePulse is a React and Next.js SaaS analytics dashboard for monitoring recurring revenue and subscription activity: monthly recurring revenue, revenue trends, subscription analytics, customer metrics, retention, cohort analysis, and account management. As the dataset grew, the dashboard began showing measurable rendering delays across its core interactions.

THE PROBLEM

Users experienced slow KPI card loading, delayed revenue chart rendering, laggy subscription search, slow plan filtering, and a heavy dashboard refresh. An initial React Profiler recording measured a render time of 2000ms.

ROOT CAUSE ANALYSIS

- KPI rendering was artificially delayed before the dashboard became interactive.
- The revenue chart waited before rendering, delaying access to historical data.
- Changing the subscription plan filter triggered a full dashboard-wide refresh instead of a scoped update.
- Every search keystroke re-filtered the full subscription dataset with no memoization, competing directly with rendering.

OPTIMIZATIONS IMPLEMENTED

- Removed artificial delays from KPI and chart rendering.
- Memoized revenue calculations with `useMemo` so they recompute only when the underlying data changes.
- Memoized subscription filtering so filter operations run only when search text or the selected plan changes.
- Moved search updates off the render-blocking path with `useDeferredValue`.
- Reduced table rendering to the visible subscription rows rather than the full dataset.

RESULTS

Render time improved from 2000ms to 140ms, validated with React Profiler.

Metric	Before	After
Render Time	2000ms	140ms
KPI Loading	Delayed	Immediate
Chart Loading	Slow	Fast
Search	Laggy	Smooth
Filter Response	Slow	Fast
Dashboard Responsiveness	Poor	Strong

TECHNIQUES USED

React Profiler, useMemo, useDeferredValue, memoized computation, rendering optimization, large-dataset optimization.

LINKS

Before / after optimization recordings and this document are linked from the RevenuePulse case study on the portfolio site.

Hritwik Tiwary — Frontend Engineer (React / Next.js)